

Multi-Period Compliance MFG with Deep FBSDE Solver

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ACKNOWLEDGMENT

This is an overview of the 2024 summer research supervised by Prof. Steven Campbell, who's dedication and guidance was sincerely appreciated.

For more math and algorithm details, please visit my GitHub repository the the link: <https://github.com/OrangeAoo/Multi-Period-Compliance-MFG-FBSDEs>. Should you find any problems, feel free to start an issue on my GitHub repository or directly reach out to me through email (ya2538@columbia.edu).

Abstract

This work aims to extend the single-period compliance model in [1] to multi-period, proposing several tricks to improve the numeric stability of the Deep Solver for FBSDEs with jumps. After validating the model framework by reproducing the results in [1], we consider an additional period, to which any remaining inventories after the first compliance period can be carried forward. For benchmarking, we run the single-period model for the 2 periods separately. Also, we compare different preferences of agents across sub-populations. This can reveal practical implications when making production strategies for companies in REC markets.

1 Problem Overview

Conventional numerical solvers are hard-pressed to solve PA-MFG with market-clearing conditions, which may be faced with the “curse of dimensionality” (Bellman 1957)¹. Thus in their study [1], Professor Campbell and his fellows proposed an actor-critic approach to optimization, where the agents form a Nash equilibria according to the principal’s penalty function and the principal evaluates the resulting equilibria. They apply this approach to a stylized PA problem arising in Renewable Energy Certificate (REC) markets, where agents may *work* overtime (or *rent* capacity), *trade* RECs, and *expand* their long-term capacity to navigate the market at maximum profit. Here we only discuss the agents’ problem in the multi-agent-multi-period scenario.

1.1 REC Market Basics

Closely related to carbon cap-and-trade (C&T) markets, REC markets are a type of market-based emissions regulation policies, which are motivating real-world applications of FBSDEs in modeling PA-MFG.

In RES markets, a regulator plays the role of principle, setting a floor on the amount of energy generated from renewable resources (aka. green energy) for each firm (based on a percentage of their total production), and providing certificates for each MWh of green energy generated and delivered to the grid. These certificates can be further traded by individual or companies, i.e. agents, to 1) reduce costs or the greenhouse gas (GHG) emissions impact of their operations; and 2) earn profits from the extra inventories instead of wasting. Since the certificates are traded assets, energy suppliers can trade-off between producing clean electricity themselves, and purchasing the certificates on the market. In all, such policies have played an important role in funding clean energy development, particularly in past years when the cost of green power production was not as competitive with the cost of fossil fuel power.

To ensure compliance, each firm must surrender RECs totaling the floor at the end of a compliance period, with a monetary penalty paid for each lacking certificate. In practice, these systems regulate multiple consecutive and disjoint compliance periods, which are linked together through a mechanism called *banking*, where unused allowances in current period can be carried on to the next period (or multiple future periods). Thus, as an extension to the single-period framework [1], we now consider a 2-period model in this report.²

¹Bellman, R. E.: Dynamic Programming. Princeton University Press, USA (1957).

²At a finite set of joint points, the possible lack of differentiability will not have any significant effects.

1.2 REC Market Modeling with FBSDEs

Let's consider 2 sub-populations here. Before jumping into the 2-period case, we first reproduce the single-period case following steps in [1]. We denote the period end as T , which can be thought of "the end of the world". Referring to the probabilistic method in [2] (R. Carmona, F. Delarue, 2012), one can show that, for agent i in sub-population k , the optimal solution to its problem in a *single* period is exactly the solution to the following coupled FBSDEs:

Now consider the 2-agent-2-period MFG with market-clearing conditions. Let's denote the 2 compliance periods $[0, T_1]$ and $(T_1, T_2]$ as $\mathfrak{T}_1$ and $\mathfrak{T}_2$, respectively. Here we think of T_2 as "the end of the world", after which there are no costs occurs and all agents forfeit any remaining RECs. Similarly, one can prove that the optimal operation for agent i in sub-population k ($\forall i \in \mathfrak{N}_k, k \in \{1, 2\}$) can be modeled with the following coupled FBSDEs:

$$\begin{cases} dX_t^i = (h^k + g_t^i + \Gamma_t^i + C_t^i)dt + \sigma^k dW_t^k - \min(X_{T_1}^i, K) \mathbf{1}_{t=T_1}, & X_0^i = \zeta^i \sim \mathcal{N}(v^k, \eta^k) \\ dC_t^i = a_t^i dt, & C_0^k = 0 \\ dV_t^i = Z_t^{V,k} dW_t^i, & V_{T_1}^i = w * \mathbf{1}_{X_{T_1}^i < K} \\ dU_t^i = Z_t^{U,k} dW_t^i, & U_{T_1}^i = 1 * Y_{T_1}^i \mathbf{1}_{X_{T_1}^i > K} \\ dY_t^i = Z_t^{Y,k} dW_t^i, & Y_{T_2}^i = w * \mathbf{1}_{X_{T_2}^i < K} \end{cases} \quad (1)$$

where the optimal controls are given by: (2)

$$S_t = \frac{\sum_{k \in \mathcal{K}} \frac{\pi^k}{\gamma^k} \mathbb{E}[V_t^i + U_t^i \mid i \in \mathfrak{N}^k; \mathcal{F}_t]}{\sum_{k \in \mathcal{K}} (\pi^k / \gamma^k)} \mathbf{1}_{t \in [0, T_1]} + \frac{\sum_{k \in \mathcal{K}} \frac{\pi^k}{\gamma^k} \mathbb{E}[Y_t^i \mid i \in \mathfrak{N}^k; \mathcal{F}_t]}{\sum_{k \in \mathcal{K}} (\pi^k / \gamma^k)} \mathbf{1}_{t \in (T_1, T_2]} \quad (3)$$

$$g_t^i = \frac{V_t^i + U_t^i}{\zeta^k} \mathbf{1}_{t \in [0, T_1]} + \frac{Y_t^i}{\zeta^k} \mathbf{1}_{t \in (T_1, T_2]} \quad (4)$$

$$\Gamma_t^i = \frac{V_t^i + U_t^i - S_t}{\gamma^k} \mathbf{1}_{t \in [0, T_1]} + \frac{Y_t^i - S_t}{\gamma^k} \mathbf{1}_{t \in (T_1, T_2]} \quad (5)$$

$$a_t^i = \frac{(T_1 - t)(V_t^i + U_t^i) + (T_2 - T_1)Y_t^i}{\beta^k} \mathbf{1}_{t \in [0, T_1]} + \frac{(T_2 - t)Y_t^i}{\beta^k} \mathbf{1}_{t \in (T_1, T_2]} \quad (6)$$

$$(7)$$

The parameters for each sub-population are given in the following table:

	π^k	h^k	σ^k	ζ^k	γ^k	v^k	η^k	β^k
k=1	0.25	0.2	0.1	1.75	1.25	0.6	0.1	1.0
k=2	0.75	0.5	0.15	1.25	1.75	0.2	0.1	1.0

Table 1: Sub-Population Parameter Values

The framework above can be extended to more realistic models with more than 2 sub-populations and compliance periods, with a penalty approximated by multi-knot functions.

2 Algorithm And Numeric Tricks

2.1 Algorithms: Joint-Optimization Vs. Separate-Optimization

To solve the said FBSDEs in 1.2, we implement the *shooting method* with *Deep Solvers* [3], discretizing the SDEs in a fine time grid and parameterizing the co-adjoint processes and initial values with neural nets. Let $\mathfrak{T} = \{t_0, \dots, t_m\}$ be a discrete set of points with $t_0 = 0$ and $T_m = T$, where m is the number of time steps. Here the step size $dt = (t_i - t_{i-1})$ is a constant and $dt = T/m$. The smaller the value of h , the closer our discretized paths will be to the continuous-time paths we wish to simulate. Certainly, this will be at the expense of greater computational effort. While there are plenty of discretization schemes available, the simplest and most common scheme is the *Euler scheme*, which is intuitive and easy to implement. In particular, it satisfies the *practical decision-making process* - make decisions for the next point of time conditioned on the current information.

The aforementioned *shooting method* is implemented by the *stepwise approximation*: starting from the initial conditions and “shoot” for the “correct” terminal conditions - the “correctness” of terminal approximations will be evaluated by computing the aggregated average forward loss/error over the whole population against corresponding targets (denoted as $\mathcal{L}$). For instance, for the single-period case, the aggregated average forward MSE after m iterations is computed as:

$$\mathcal{L}(\theta^{(m)}) = \sum_{i \in \mathfrak{N}} (Y_T^i - w \mathbf{1}_{X_T^i < K})^2,$$

and for the 2-period case:

$$\mathcal{L}(\theta^{(m)}) = \sum_{i \in \mathfrak{N}} (V_{T_1}^i - w \mathbf{1}_{X_{T_1}^i < K})^2 + \sum_{i \in \mathfrak{N}} (U_{T_1}^i - Y_{T_1}^i \mathbf{1}_{X_{T_1}^i > K})^2 + \sum_{i \in \mathfrak{N}} (Y_{T_2}^i - w \mathbf{1}_{X_{T_2}^i < K})^2.$$

The algorithm takes major steps as follows:

1. start from the neural nets for initial values (i.e. Y_0^i etc.);
2. compute the process values at every time step;
3. get approximations to terminal conditions and compute $\mathcal{L}$;
4. compute gradients of $\mathcal{L}$ against parameters (weights and biases, denoted as $\theta^{(m)}$) in the neural nets (i.e. Y_0^i and Z_t^k etc.) and take gradient steps to determine the next set of parameters.

More specifically, the above steps can be more explicitly displayed by the pseudocode in the [Appendix](#) section.

To benchmark the jointly optimized 2-period Model, we first run the 1-period algorithm for each period, i.e. minimize the agents’ costs in either period separately.

2.2 Numeric Tricks

The trickiest problem we are facing is the indicator functions in *terminal conditions*. Another problem is that the ordinary Deep FBSDE Solver may learn $V_t^i, U_t^i, Y_t^i \notin [0, 1]$ (let’s fix $w = 1$ for now), which is meaningless as they represent the *probabilities* of defaulting (i.e. missing the quota). To solve them, we

propose 3 numeric tricks and integrate a combined method into the algorithm.

Sigmoid Approximation A natural way to increase continuity and differentiability is by (see Figure 6 in the [Appendix](#) section):

$$\mathbf{1}_{0.9 > x} \approx \sigma(0.9 - x), \text{ where the sigmoid function } \sigma(u) = \frac{1}{1 + e^{-u/\delta}}. \quad (8)$$

In particular, the parameter δ controls the steepness of $\sigma(\cdot)$ and usually is a small positive number - the smaller δ is, the more closely it approximates the step of the indicator function.

Numeric Clamp Instead of using `tensor.clamp` to brutally clamp values within $[0, 1]$, we introduce a more numerically stable way, especially for values close to the 2 interval ends (same applies to V_t^i, U_t^i).

$$dY_t^i = Y_t^i(1 - Y_t^i)Z_t dB_t. \quad (9)$$

Nonetheless, there are limitations to the above approaches. For the *sigmoid approximation*, when δ is too small, there is a great potential for numerical overflow - the exponents could be tremendous especially when X_t is far greater than 0.9, such that `torch.exp(u)` is `inf` when $u \geq 7.1$. This will raise errors/warnings³ in PyTorch. And for *numeric clamp* to work, we must ensure the **initial values** strictly fall in $(0, 1)$. Thus we propose the third approach:

Logit-Clamp Transformation To map the range $[0, 1] \rightarrow \mathbb{R}$ while avoids working with large exponents, we let: $\tilde{Y} := w * \text{logit}(Y/w)$. Then apply *Itô's formula* (with superscript $[\cdot]^i$ omitted):

$$d\tilde{Y}_t = (w/2 - Y_t)Z_t^2 dt + wZ_t dB_t. \quad (10)$$

Correspondingly, we use [BCEWithLogitsLoss](#) as the loss function, which combines a Sigmoid layer and the [BCELoss](#) in one single class. This version is more numerically stable than using a plain *Sigmoid* followed by a [BCELoss](#) as, by combining the operations into one layer, it takes advantage of the log-sum-exp trick for numerical stability.

Worth mentioning, we experimented with multiple combinations of tricks and loss functions, paired with different optimizers and schedulers. Eventually, we chose [Adamax](#) and [StepLR](#) due to their relatively better and more stable performance for all cases in general. Specifically, the 4 valid combinations of tricks and loss functions are shown as follows.

	target type	trick	loss type
1	indicator	logit	BCEWithLogitsLoss
2	indicator	clamp	BCELoss
3	indicator	clamp	MSELoss
4	sigmoid	clamp	MSELoss

Table 2: Valid Combos

³Examples of [RuntimeError](#) and [RuntimeWarning](#) on PyTorch Forums.

3 Results

To facilitate the evaluation of algorithm performances, `plot_results` - a user-friendly class - is constructed.⁴, which also visualizes agents' behaviors (or interactions) and their market impacts. Specifically, it produces results from the following aspects:

- **Agents' behaviors and market impacts**
 - Learnt optimal control processes
 - Decomposed inventory accumulation processes
 - Inventory levels during 2 compliance periods
 - Terminal inventories ready-to-submit
 - Market-clearing prices
- **Algorithm convergency and learning loss**
 - Average forward losses against a number of epochs trained
 - Learnt terminal conditions vs. targets

And here are some example diagrams.

3.1 Jointly Optimized 2-Agent-2-Period Sample Results

Here we use the sample results by model⁵ with terminal target function: $0.25\sigma(0.9 - X_T^i)$, where $\delta = 0.03$ and $T = T_1, T_2$. The loss function is *MSELoss* and the trick used in the *stepwise approximation* is *clamp*, correspondingly. (See Table 2.)

The learned controls are shown as the first row of Figure 1 and the accumulative generations by different means as the second row, correspondingly. The green color denotes the sub-population 1 while red the sub-population 2. (The same applies to all the later plots.) The plots in Figure 5 display a rather good convergence of learned terminal values to their *sigmoid targets*.

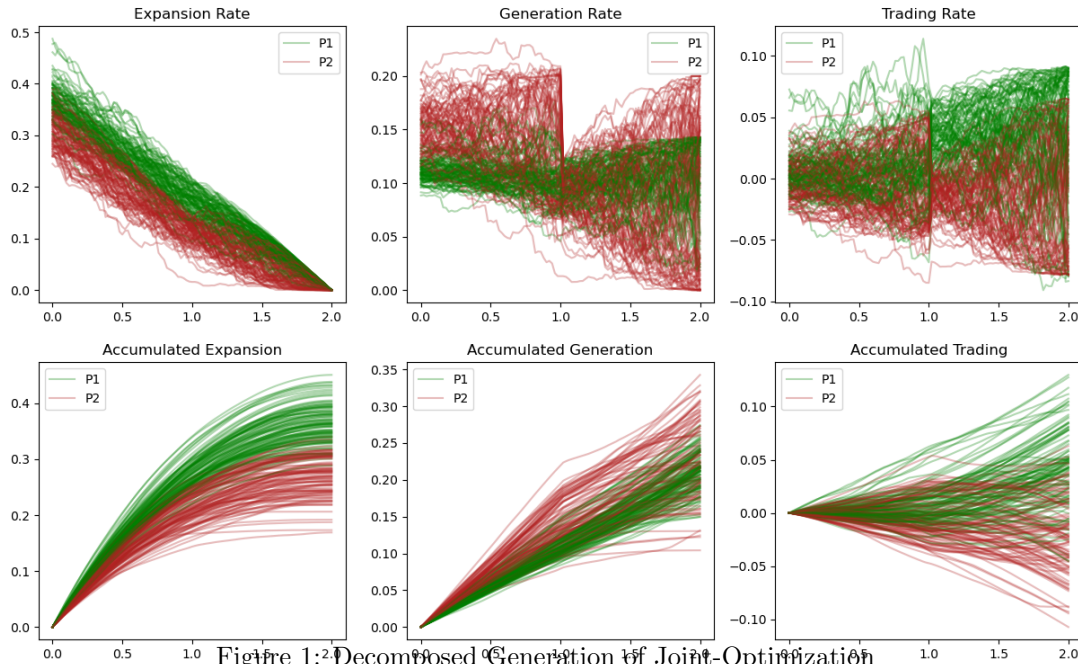


Figure 1: Decomposed Generation of Joint-Optimization

⁴See more instruction details in GitHub [README](#) for the Joint-Optimization or Separate-Optimization.

⁵The code and full results can be found in the [GitHub Repository](#).

3.2 Separately Optimized 2-Agent-2-Period Sample Results

For the sake of comparability, here we use the same numeric trick, target, and loss function, except for minor changes when fine-tuning the specific model parameters⁶.

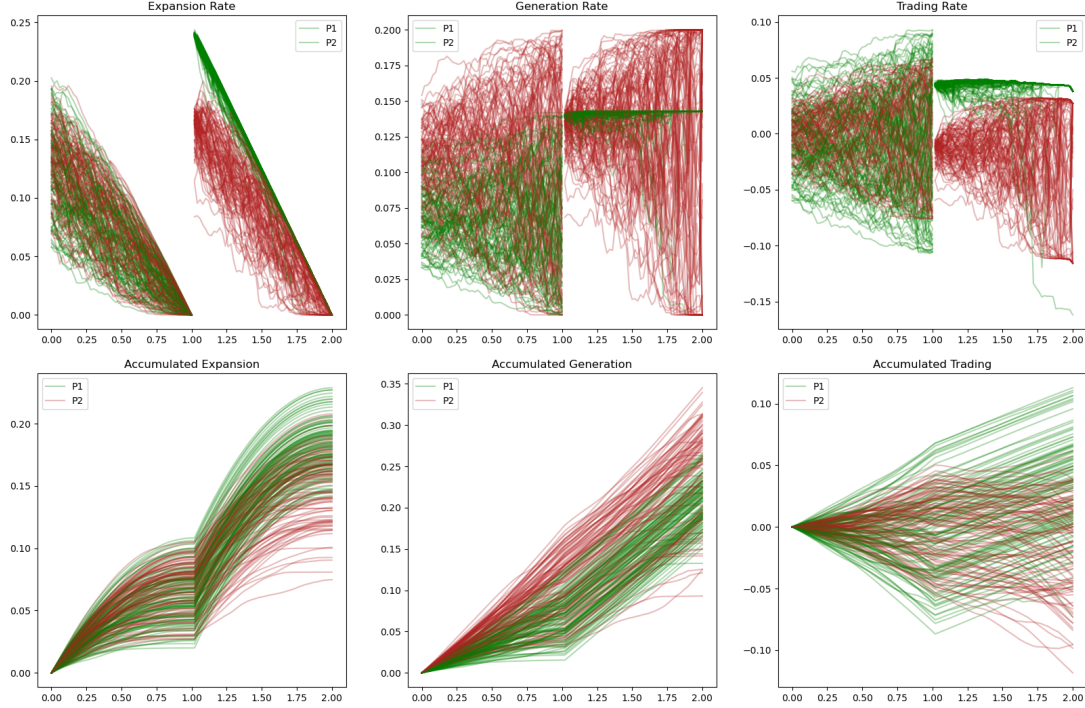


Figure 2: Decomposed Generation of Separate-Optimization

3.3 Comparisons And Analyses

Joint Vs Separate Optimization Upon comparing the shown results from 2 different perspectives, one can get enlightening implications.

In contrast to the short-sighted agents, those who plan ahead in $\mathfrak{T}_1$ tend to invest more in expansion even at the end of the first period, generating slightly more than the quota required.

Therefore in $\mathfrak{T}_2$, they are saved from starting from scratch. And since most of them have already accumulated a relatively high baseline rate, there's less need to work overtime or purchase inventories than in the first period (shown by the decreased slopes of the accumulated generation plots). However, the agents in the benchmark model have to either 1) try very hard to make up for expansion (exemplified by the green “pop1”), or 2) rely heavily on over-hours and trading (exemplified by the red “pop2”).

Their market impacts are reflected in the trading prices: the greater the demand, the higher the prices. This is depicted by Figure 3.

⁶The code and full results can be found in the [GitHub Repository](#).

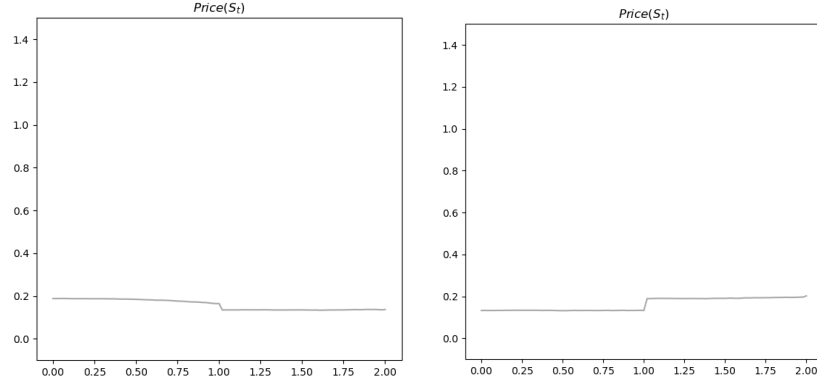


Figure 3: Market Prices: Joint (L) Vs Separate (R) Optimization

Sub-Population 1 Vs 2 A closer look at each sub-population reveals differences due to distinct “preferences” and initial conditions. Regardless of perspective (long/short-term), the higher initial inventory of “pop1” (green) makes them less motivated to work extra hours in the first period, despite a hidden drawback in baseline capacity. As a result, in the second compliance period, they often struggle to meet the quota and must purchase from “pop2” (red), as shown by the red uptrends and green downtrends in Figures 1 and 2 (where positive values indicate buying and vice versa).

Model Performances As is seen from Figure 4 and 5, the algorithms converge quickly after 200 epochs, and their learned terminal conditions align closely with target values (displayed as black dots).

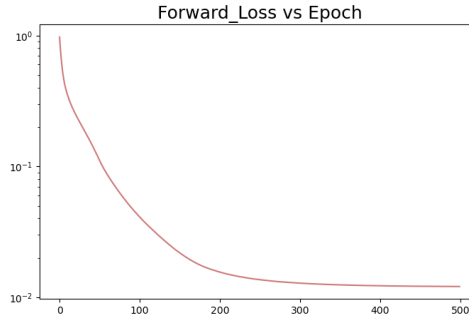
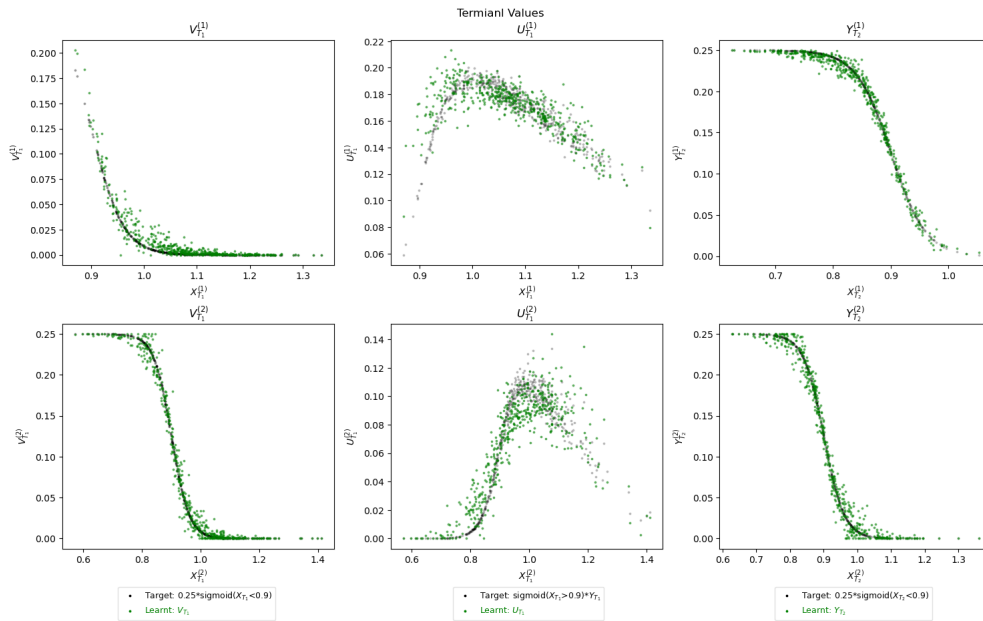


Figure 4: Forward Loss

Figure 5: Terminal Values of V, U, Y

4 Conclusions And Takeaways

In this study, we successfully extended the single-period compliance model to a 2-period setting, providing a motivating framework for modeling REC markets using coupled FBSDEs.

Methodologically, the additional smoothing and scaling novel numeric techniques including *sigmoid approximation*, *numeric clamp*, and *logit-clamp transformation*, contributed to improved model stability and convergence.

Practically, the results highlighted key differences between agent preferences across sub-populations. Additionally, by benchmarking joint optimization to a separate one, we revealed significant insights into agent behaviors and market prices influenced by long-term versus myopia perspectives on compliance strategies, giving practical and instructive takeaways not only to the REC markets but also to one's life.

- *Always plan ahead and consider for the future.*
- *Always do slightly more than required and maintain a reasonable level of backups.*
- *Don't be blinded by the apparent advantages/achievements, instead care for the growth rate and capacity - that's what you can carry to the future for sure.*
- *When the majority gets lazy or short-sighted, the market gets worse - where any individual will be affected more or less.*

This work sets a foundation for further exploration into higher-dimensional MFG FBSDEs and more realistic REC scenarios. **In the next phase, one is encouraged to add more compliance periods and/or inventory carryforward (or expiration) rules.** And to facilitate the experiments, the models are structured as user-friendly packages, documented by comprehensive instructions, and made public in [my GitHub repository](#), which is open to suggestions and/or contributions.

Appendix

FBSDEs For The Single-Period-2-Agent Model

$$\begin{cases} dX_t^i = (h^k + g_t^i + \Gamma_t^i + C_t^i)dt + \sigma^k dW_t^k, & X_0^i \sim \mathcal{N}(v^k, \eta^k) \\ dC_t^i = a_t^i dt, & C_0^i = 0 \\ dY_t^i = Z_t^k dW_t^k, & Y_T^i = w \mathbf{1}_{X_T^i < K}, \end{cases} \quad (11)$$

where:

$$\begin{aligned} Y_t^i &= \mathbb{E} \left[w \mathbf{1}_{X_T^i < K} | \mathcal{F}_t \right] = w \mathbb{P} (X_T^i < K | \mathcal{F}_t) \\ S_t &= \frac{\sum_{k \in \mathcal{K}} \left(\frac{\pi^k}{\gamma^k} \mathbb{E} [Y_t^i | i \in \mathfrak{N}^k; \mathcal{F}_t] \right)}{\sum_{k \in \mathcal{K}} \left(\frac{\pi^k}{\gamma^k} \right)} \\ g_t^k &= \frac{Y_t^k}{\zeta^k} \\ \Gamma_t^k &= \frac{Y_t^k - S_t}{\gamma^k} \end{aligned} \quad (12)$$

Key Notations

The key notations/parameters in the FBSDEs are interpreted as follows:

- $k \in \mathcal{K}$: a sub-population of agents, within which all individuals are assumed to have identical preferences and similar initial conditions/capacities, yet across which are distinct. The sub-population is annotated by superscript $[\cdot]^k$. Here we only discuss $k = 1, 2$.
- $i \in \mathfrak{N}$: an individual agent belonging to the sub-population $\mathfrak{N}^k$, annotated by superscript $[\cdot]^i$.
- $X_t := (X_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the current inventories in stock. For some key time points:
 - at $t = 0$, there may be some stochastics in the initial inventories, which are assumed to be normally distributed. $X_0^i \sim \mathcal{N}(v^k, \eta^k)$, $\forall k \in \mathcal{K}$, $\forall i \in \mathfrak{N}^k$.
 - at $t = T_1$, the terminal RECs pre-submission are X_{T_1} carried over from the first period. Shortly after forfeiting $\min(K, X_{T_1}^i)$, the remaining inventories in stock are $ReLU(X_{T_1}^i - K)$, which are treated as new initial values for the second period.
 - at $t = T_2$, the terminal RECs pre-submission are $X_{T_2}^i$.
- $I_t := (I_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the integrated inventory generation. We introduce this process for continuous differentiability at T_1 . And X_t has the same initial conditions as I_t . We have:

$$X_t = \begin{cases} I_t, & t \in [0, T_1] \\ I_t - \min(I_{T_1}, K), & t \in (T_1, T_2] \end{cases} \quad \text{or} \quad X_t = \begin{cases} I_t, & t \in [0, T_1] \\ I_t - I_{T_1} + (I_{T_1} - K)_+, & t \in (T_1, T_2]. \end{cases} \quad (13)$$

- K : the quota that agents must meet at the end of each compliance period. Fixed to $K = 0.9$ ⁷.
- $P(\cdot)$: the generic penalty function approximated by *single-knot penalty functions*⁸ :

$$P(x) = w(0.9 - x)_+ \Rightarrow \partial_x P(x) = -w \mathbf{1}_{x < K}.$$

Further, by tuning the weight w , we can see the relation between the penalty level (controlled by w) and the agents' behaviour, as well as its market impact.

- h : the baseline generation rate at which agents generate inventories with zero marginal cost.
- $C_t := (C_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: incremental REC capacity of agents, i.e. increase of baseline generation rate over time, accumulated by investing in expansion plans - for instance, by installing more solar panels.⁹
- $a_t := (a_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the control of expansion rate, representing long-term REC capacity added per unit time. Note that it could be made even more realistic by incorporating a *delay* between the decision to expand (a_t) and the increase to the baseline rate h .
- $g_t := (g_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the control of overtime-generation rate, i.e. extra capacity achieved by working extra hours and/or renting short-term REC generation capacity at an assumed quadratic cost - specifically, over-hour bonus and/or rental fees.
- $\Gamma_t := (\Gamma_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the control of trading rate, with negative¹⁰ values being the amount sold whereas positive purchased per unit of time.
- $S_t := (S_t)_{t \in \mathfrak{T}_1 \cup \mathfrak{T}_2}$: the equilibrium REC price obtained endogenously through market-clearing condition:

$$\lim_{N \rightarrow \inf} \frac{1}{N} \sum_{i \in \mathfrak{N}} \Gamma_t^i = 0$$

- ζ, γ, β : scalar cost parameters which are identical for agents within the same sub-population.
- π : the proportion of each sub-population:

$$\pi^k = \frac{|\mathfrak{N}^k|}{\sum_{j \in \mathcal{K}} |\mathfrak{N}^j|}.$$

⁷The choice of knot point is associated with h^k and total time span T_1, T_2 . A good target (or quota) should be “**attainable**” - neither too easy nor too hard to achieve. Specifically, even if agents do nothing at all, they will have an initial amount plus a baseline generation of inventories - for instance, $0.2 * 1 + 0.6 = 0.8$ for agents in sub-population 1 at the first-period end. Similarly, for sub-population 2, all agents will also have a “*guaranteed*” level of 0.8 for delivery. Thus a target reasonably higher than that, i.e. 0.9, would be regarded “**attainable**”.

⁸See *Report-StepwiseDetail* for more math details.

⁹The incremental capacity over baseline can be carried forward to future periods.

¹⁰While trading rate may be positive or negative, expansion and overtime-generation rates must be positive.

Sigmoid Approximation

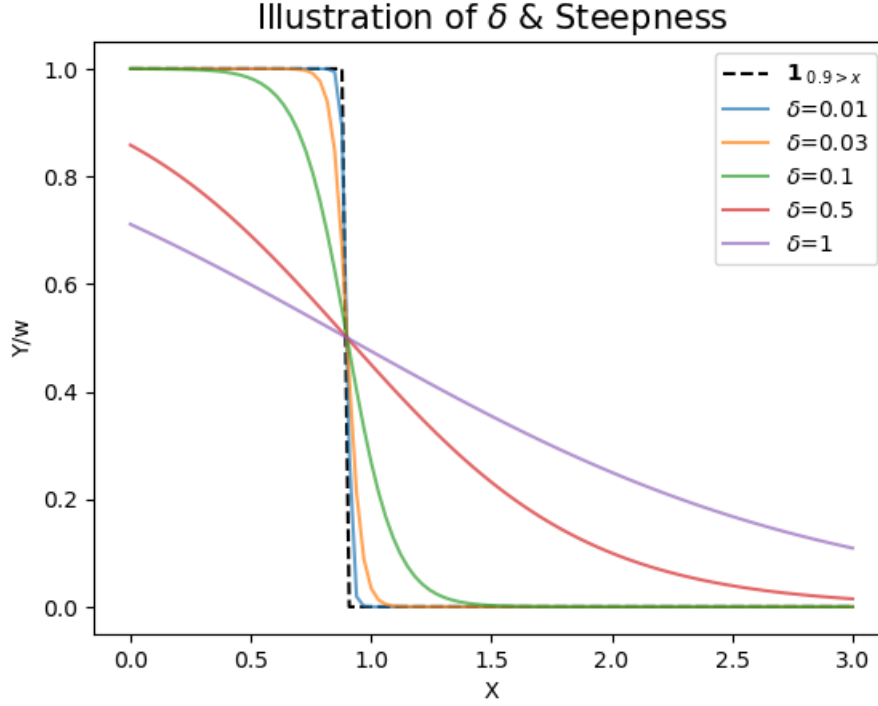


Figure 6: Sigmoid Approximation

Algorithms

The stepwise function to approximate the discretized coupled FBSDEs and the main algorithm - forward training loop - is structured correspondingly as:

Algorithm 1: Shooting Method - Stepwise Approximation

Input : Initial conditions and untrained models

Output: Aggregated forward loss ToTLoss

```

1 Function get_forward_loss(initial conditions and models):
2   Initialization;
3   for  $j \leftarrow 1$  to  $T_2$  do
4     Update  $\mathbb{P}$  (default) for each sub-population:
5      $V_j \leftarrow V_{j-1} + \text{zv\_models}[j-1] * dB_j$ ,  $U_j \leftarrow U_{j-1} + \text{zu\_models}[j-1] * dB_j$ ,  $Y_j \leftarrow Y_{j-1} +$ 
       $\text{zy\_models}[j-1] * dB_j$ ;
6     Compute overtime-generation, trading, and expansion rates:  $g$ ,  $\Gamma$ , and  $a$ ;
7     Update  $C$  and  $X$ ;
8     if  $j=T_1$  then
9       Freeze  $X$ ,  $V$ , and  $U$  at  $T_1$ ;
10      Submit inventories:  $X \leftarrow \text{ReLU}(X - K)$ ;
11    end
12  end
13  Aggregate the losses:  $\text{ToTLoss} \leftarrow \text{CalTerminalLoss}(V, U, Y)$ ;
14  return ToTLoss

```

References

- [1] S. Campbell, Y. Chen, A. Shrivats, and S. Jaimungal, *Deep learning for principal-agent mean field games*, 2021. arXiv: 2110.01127 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2110.01127>.

Algorithm 2: Training Loop for Forward Loss Approximation

Input : Sample X_0, C_0 , and BM Increments dB for each sub-population**Output**: Trajectories of $X_t, g_t, \Gamma_t, a_t, C_t$, and terminal conditions $V_{T_1}, U_{T_1}, Y_{T_2}$

```
1 begin
2   Initialization;
3   for  $batch \leftarrow 0$  to MaxBatch do
4     SumLoss  $\leftarrow 0$ ;
5     for  $iter \leftarrow 0$  to OptimSteps do
6       Compute forward loss:  $loss \leftarrow \text{get\_forward\_loss}(\text{initial conditions})$ ;
7       Backpropagate:  $loss.backward()$ ;
8       Update parameters:  $optimizer.zero\_grad(), optimizer.step()$ ;
9       Adjust learning rate:  $scheduler.step()$ ;
10      SumLoss  $\leftarrow \text{SumLoss} + loss$ ;
11    end
12    AppendToForwardLosses(SumLoss/OptimSteps);
13    if not train on the single batch then
14      | Generate a new batch with updated initial processes;
15    end
16  end
17  Visualize the results: Forward Losses,  $X_t, g_t, \Gamma_t, a_t, C_t$ , and terminal convergence.
18 end
```

- [2] R. Carmona and F. Delarue, *Probabilistic analysis of mean-field games*, 2012. arXiv: [1210.5780](https://arxiv.org/abs/1210.5780) [math.PR]. [Online]. Available: <https://arxiv.org/abs/1210.5780>.
- [3] J. Han and J. Long, "Convergence of the deep bsde method for coupled fbsdes.," *Probability, Uncertainty and Quantitative Risk*, vol. 5, 2020. [Online]. Available: <https://doi.org/10.1186/s41546-020-00047-w>.